

Job Market Analysis Using MapReduce and Apache Spark

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The objective of this project is to analyze a large job market dataset containing information about occupations, salaries, experience levels, education levels and geographic locations. The analysis aims to understand the key factors that influence salary and to identify patterns within the job market. In particular, levels of experience and educational backgrounds, as well as how job opportunities are distributed geographically. To achieve this, Big Data processing techniques are applied using both MapReduce and Apache Spark. MapReduce is used to simulate distributed data processing while Spark is used to build a complete and scalable data processing pipeline.

Through this analysis, the project seeks to provide meaningful insights into the relationship between professional characteristics and earning potential, as well as to highlight broader trends in the job market.

The dataset used in this project was obtained from Kaggle and contains job market information such as salaries, occupations, and experience levels.

Dataset source: <https://www.kaggle.com/datasets/kundanbedmutha/job-market-dataset-global>

In this project, the MapReduce framework is applied to compute the average salary for each occupation in the dataset. This task was chosen because it provides a meaningful way to compare different job roles while demonstrating the core principles of distributed data processing.

The mapper function processes each record in the dataset and transforms it into a key-value pair where the key represents the occupation and the value represents the salary. This step converts the raw data into a structured format suitable for aggregation.

During the shuffle phase, all values associated with the same key are grouped together, meaning that all salaries corresponding to a specific occupation are collected into a single group. This organization is essential for performing aggregation operations.

The reducer function then processes each group by calculating the average salary for each occupation. This is done by summing all salary values within a group and dividing by the number of records.

To better simulate a real Big Data environment, the dataset is divided into smaller chunks and processed separately. Each chunk undergoes the mapping and local grouping steps and the partial results are later combined using the reduce function. This approach reflects how distributed systems process large datasets in parallel across multiple machines.

Overall, this implementation demonstrates how MapReduce can efficiently handle large-scale data processing by breaking down tasks into smaller operations and aggregating results in a structured manner.

The Apache Spark workflow begins with initializing a Spark session and loading the dataset into a Spark DataFrame. The structure of the dataset is then explored by examining the schema, previewing sample records and identifying the number of rows and columns.

Data processing is performed by applying filtering conditions to focus on relevant subsets of the data and selecting only the most important columns for analysis. Feature engineering is then used to enhance the dataset by creating new variables such as salary categories and experience levels, which simplify analysis and improve interpretability.

Further transformations are applied through advanced operations, including sorting and aggregations. These operations allow for the computation of key metrics such as average salary, minimum and maximum values and job counts across different groups.

Exploratory data analysis is conducted to identify patterns and trends in the dataset. This includes analyzing job distribution across countries and examining how salary varies across different categories.

To improve performance, optimization techniques are applied by reducing the number of columns in the dataset, which decreases memory usage and enhances processing efficiency. Finally, the dataset is prepared for machine learning by splitting it into training and testing sets demonstrating how the processed data can be used for future predictive tasks.

The analysis of the dataset reveals several important patterns related to salary distribution in the job market.

One of the most significant findings is that salary varies greatly across occupations. Technical roles such as AI Engineer, Cloud Engineer and Data Scientist consistently rank among the highest-paying positions, indicating that advanced technical skills are highly valued. In addition, managerial roles such as Product Manager and Operations Manager also show high salary levels, suggesting that leadership and strategic responsibilities play a key role in determining compensation.

Experience level is another major factor influencing salary. The results show a clear progression in earnings from junior to mid-level and senior positions, with senior professionals earning significantly higher salaries. This highlights the importance of career development and accumulated experience in achieving higher income levels.

Education level also impacts salary, particularly at higher levels of experience. Individuals with advanced degrees such as Master's and PhD, tend to earn more, especially in senior roles. However, at earlier career stages, the difference between education levels is less pronounced suggesting that experience may have a stronger influence than education at the beginning of a career.

The analysis also reveals that the dataset is skewed toward medium and high-paying jobs, with relatively fewer low-paying positions. This suggests that the dataset may be biased toward skilled or experienced professionals.

Finally, geographical analysis shows that job opportunities are not evenly distributed across countries, with certain regions, particularly the United States, having a significantly higher number of job entries. This indicates that regional factors may influence both job availability and salary levels.

Overall, the findings demonstrate that salary is influenced by a combination of occupation, experience, education and geographic location, with experience and specialization emerging as the most consistent drivers of higher earnings.